

Dongfang Zhao, Ph.D.

Website: faculty.washington.edu/dzhao

Professional Preparation

Katholieke Universiteit Leuven, Belgium	Mathematical Statistics, Diploma	2005
Emory University, Atlanta, GA	Computer Science, MS	2008
Illinois Institute of Technology, Chicago	Computer Science, PhD	2015
University of Washington (UW), Seattle	Computer Science, PostDoc	2017

Employment

2023–present	Assistant Professor, UW Tacoma School of Engineering & Technology
2020–2023	Lead Developer, Moka Blox LLC, Davis, CA
2018–2020	Visiting Professor, University of California, Davis (UC Davis)
2018–2023	Assistant Professor, University of Nevada, Reno (UNR)
2015	Research Intern, IBM Almaden Research Center, San Jose, CA
2014	Research Intern, Argonne National Laboratory, Chicago, IL
2013	Research Intern, Pacific Northwest National Laboratory, Seattle, WA
2009–2011	Software Developer, Epic Systems Corporation, Madison, WI
2003–2004	Research Intern, Interuniversity Microelectronics Centre, Belgium

Selected Awards

2022	Federal Research Grant, UNR PI, NSF 2112345, \$255,916 “A DLT Machine Learning Platform for Blockchain Warehousing”
2020	Federal Research Grant, UNR PI, DOE SC0020455, \$200,000 “HPChain: Energy-Efficient Scalable Blockchains for HPC Systems”
2019	Best Paper Award, International Conference on Cloud Computing “BlockLite: Toward Accurate and Efficient Emulation of Public Blockchains in the Cloud”
2018	Best Student Paper Award, IEEE International Conference on Cloud Computing “Toward Cost-Effective Memory Scaling in Clouds: Symbiosis of Virtual and Physical Memory”
2016	Postdoctoral Fellowship, Sloan Foundation, \$155,000

Selected Services

2023–present	Associate Editor	Journal of Big Data (Springer Nature)
2021–present	Editorial Board	IEEE Transactions on Distributed & Parallel Systems
2018–2022	Grant Review Panels	NSF CISE IIS, NSF CISE OAC, DOE ASCR
2022	Search Committee	One Associate Professor of Databases, UNR
2022	Search Committee	One Assistant Professor of Artificial Intelligence, UNR
2021	Search Committee	Two Assistant Professors of Cybersecurity, UNR
2019	Organizing Committee	ACM International Middleware Conference, UC Davis

Selected 10 Publications

(**Dongfang Zhao** is the student's *primary advisor, [†]thesis committee member, or [‡]postdoc co-author)

- [1] *Olamide Timothy Tawose, Jun Dai, Lei Yang, **Dongfang Zhao**. Toward Efficient Homomorphic Encryption for Outsourced Databases through Parallel Caching. Proceedings of the ACM on Management of Data, SIGMOD 2023.
- [2] *Xinying Wang, Cong Xu, Ke Wang, Feng Yan, **Dongfang Zhao**. Memory Scaling of Cloud-based Big Data Systems: A Hybrid Approach. IEEE Transactions on Big Data. 8(5): 1259-1272 (2022).
- [3] *Abdullah Al-Mamun, Feng Yan, **Dongfang Zhao**. BAASH: lightweight, efficient, and reliable blockchain-as-a-service for HPC systems. The International Conference for High Performance Computing, Networking, Storage and Analysis (SC), 2021. Direct acceptance without revision
- [4] *Xinying Wang, *Olamide Timothy Tawose, Feng Yan, **Dongfang Zhao**. HDK: Toward High-Performance Deep-Learning-Based Kirchhoff Analysis. The Thirty-Fourth AAAI Conference on Artificial Intelligence (AAAI), 2020. Oral presentation
- [5] **Dongfang Zhao**, Mohamed Mohamed, Heiko Ludwig. Locality-Aware Scheduling for Containers in Cloud Computing. IEEE Transactions on Cloud Computing. 8(2): 635-646 (2020).
- [6] [†]Heyang Qin, [†]Syed Zawad, Yanqi Zhou, Lei Yang, **Dongfang Zhao**, Feng Yan. Swift machine learning model serving scheduling: a region based reinforcement learning approach. The International Conf. for High Performance Computing, Networking, Storage and Analysis (SC), 2019.
- [7] [‡]Parmita Mehta, [‡]Sven Dorkenwald, **Dongfang Zhao**, Tomer Kaftan, Alvin Cheung, Magdalena Balazinska, Ariel Rokem, Andrew J. Connolly, Jacob VanderPlas, Yusra AlSayyad. Comparative Evaluation of Big-Data Systems on Scientific Image Analytics Workloads. Proceedings of the VLDB Endowment, VLDB 2017.
- [8] **Dongfang Zhao**, Kan Qiao, Zhou Zhou, Tonglin Li, Zhihan Lu, Xiaohua Xu. Toward Efficient and Flexible Metadata Indexing of Big Data Systems. IEEE Transactions on Big Data. 3(1): 107-117 (2017).
- [9] **Dongfang Zhao**, Kan Qiao, Jian Yin, Ioan Raicu. Dynamic Virtual Chunks: On Supporting Efficient Accesses to Compressed Scientific Data. IEEE Transactions on Services Computing. 9(1): 96-109 (2016).
- [10] **Dongfang Zhao**, Ning Liu, Dries Kimpe, Robert B. Ross, Xian-He Sun, Ioan Raicu. Towards Exploring Data-Intensive Scientific Applications at Extreme Scales through Systems and Simulations. IEEE Transactions on Parallel and Distributed Systems. 27(6): 1824-1837 (2016).

Doctoral Students (name, duration, first employment)

Olamide Timothy Tawose	2018 – 2023	Assistant Professor, Lincoln University, PA
Abdullah Al-Mamun	2018 – 2022	Assistant Professor, Augusta University, GA
Xinying Wang	2018 – 2022	Research Scientist, Oak Ridge National Lab, TN

Status

U.S. citizen, married with two children